

Accelerating PageRank with Dynamic Momentum Power Iteration

*A methodological extension and cross-domain application of
Austin, Pollock & Zhu (2024)*

Group Details

Section: C

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Base Paper Overview

Austin, C., Pollock, S., & Zhu, Y. (2024). *Dynamically accelerating the power iteration with momentum.*

Numerical Linear Algebra with Applications, 31(6), e2584. DOI: 10.1002/nla.2584

What the Paper Does

- Proposes a dynamic momentum method to accelerate the classical power iteration for finding a matrix's dominant eigenpair
- Momentum parameter β_k is updated every iteration using only the Rayleigh quotient and residual ratio — no prior knowledge of λ_2 required
- Costs a single matrix-vector multiply per iteration, same as plain power iteration
- Proven convergent, with acceleration guaranteed for symmetric matrices
- Benchmarked only on abstract sparse test matrices (SuiteSparse collection) — never applied to a real-world ranking problem

Core Update Rule

$$\beta_k = \frac{\mathbf{v}_k^2 \cdot \mathbf{r}_k^2}{4}$$
$$\mathbf{u}_{k+1} = A \mathbf{x}_k - (\beta_k / h_k) \mathbf{x}_{k-1}$$

$$\rho_k = \min(d_{k+1}/d_k, 1), \quad r_{k+1} = 2\rho_k / (1 + \rho_k^2)$$

$\mathbf{v}_k$ Rayleigh quotient (eigenvalue estimate)

ρ_k Detected residual convergence rate

β_k Self-tuning momentum parameter

Problem Statement & Scope

The base paper never tests its method on a real ranking or graph problem — only synthetic and sparse benchmark matrices.

The Problem

PageRank requires finding the dominant eigenvector of a huge, sparse Google matrix — a computation that becomes slow when the two largest eigenvalues are close together (a small spectral gap), which is common in real web and citation graphs. Standard power iteration converges slowly in exactly this regime.

Our Approach

We implement the dynamic momentum method and apply it to real PageRank computation on citation and web graph datasets, benchmarking it against plain power iteration and the static-optimal momentum method to measure how much faster ranking converges in practice.

Scope of Work

- Build the PageRank / Google matrix from a real citation or web-graph dataset
- Benchmark convergence speed, iteration count, and matrix-vector multiplies across methods
- Study how the damping factor and spectral gap affect the achievable speedup

Expected Methodology

01

Implement Baselines

Plain power iteration, static-optimal momentum ($\beta = \lambda_2^2/4$), and the dynamic momentum method (Algorithm 3.1) in Python / NumPy.

02

Build the PageRank Matrix

Construct the Google matrix G from a real graph dataset (citation / web network) with damping factor $d = 0.85$, handling dangling nodes.

03

Run & Benchmark

Compare iteration count, matrix-vector multiplies, and wall-clock time to convergence across all three methods on the same dataset.

04

Extend & Analyze

Vary damping factor d and test multiple graphs; correlate speedup with the spectral gap $r = |\lambda_2/\lambda_1|$, validating the paper's theory.

Tools: Python, NumPy, SciPy (validation), NetworkX / SNAP datasets, Matplotlib for convergence plots.